

50 Exclusive Numerical Questions for PMP

1. If the Estimate at completion (EAC) is \$6500, Budget at completion is \$5500 and Estimate to complete (ETC) is \$1200, what is the variance at completion?
 - A. -1000
 - B. 1000
 - C. 100
 - D. -100

Explanation:

The correct answer is option A. $VAC = BAC - EAC = 5500 - 6500$.

2. You accept project costs to date and assume future cost variances to be atypical. Find EAC if $BAC = \$82500$, $ETC = \$30000$, $PV = \$32500$, $AC = \$20000$, $EV = \$25000$ and $Cum\ CPI = 1.25$.
 - A. \$77,500
 - B. \$66,500
 - C. \$66,000
 - D. \$65,500

Explanation:

The correct answer is option A

$$\begin{aligned} EAC &= AC + BAC - EV \\ &= 20000 + 82500 - 25000 \\ &= \$77,500 \end{aligned}$$

3. Following project data are given to you:

- a) Target price = \$90000
- b) Ceiling price = \$100000
- c) Customer's share of cost overruns = 70 %
- d) Target cost = \$80000

You are requested to find out point at which the seller will assume all subsequent costs:

- A. \$ 104286
- B. \$ 80000
- C. \$ 14286
- D. \$ 94286

Explanation:

Choice D is the correct answer. Point of Total Assumption (PTA) is a contract clause designed to protect the customer from being charged for costs going beyond a certain threshold limit.

$$PTA = [(Ceiling\ price - target\ price) / customer's\ share\ of\ cost\ overruns] + target\ cost$$

$$PTA = ((100000 - 90000) / 70\%) + \$80000$$

$$PTA = \$94286$$

4. You have been given a project that is having a Time & Material (T&M) type of contract signed. The service provider initially estimates that the total effort involved would be about 1500 hours of total effort. The project is contracted at a rate of US\$ 60 per hour.

If the project ended up with 1750 hours of total effort, what would the final contract payment would be?

- A. US\$ 90,000
- B. US\$ 1,05,000
- C. US\$ 75,000
- D. US\$ 120,000

Explanation:

Choice B is the correct answer. The correct answer is US\$ 1,05,000.

Since this is a T&M contract, the contract payment is open-ended in value.

Hence the contract payment value is the actual effort multiplied by the agreed rate

$$= \text{actual effort} \times \text{agreed rate}$$

$$= US\$ 60 \times 1750$$

$$= US\$ 1,05,000$$

5. In a project the following data was provided by one of your team leader, BAC=\$500,000, PV=\$325,000, AC=\$275,000 and cum EV=\$250,000, you are experiencing typical variances, find ETC.

- A. \$ 265,000
- B. \$ 275,000
- C. \$ 278,000
- D. \$ 290,000

Explanation:

Choice B is the correct answer

$$ETC = BAC - \text{cumEV} / CPI \text{ (cum stands for cumulative)}$$

$$CPI = 250 / 275$$

$$ETC = (500K - 250K) / (250 / 275) = 275K$$

6. Your SVP has asked you to calculate the Estimate At Completion is going to be for very small project you are working on. You were given a budget of \$3000, and to date you have spent \$2000 but only completed \$1000 worth of work.
- A. \$ 4,000
 - B. \$ 5,000
 - C. \$ 6,000
 - D. Either A or C

Explanation:

Choice D is the correct answer. You can quote both the answers with the scenario of typical and atypical variance depending on the project progress.

<i>Solution 1</i>	<i>Solution 2</i>
$EAC = AC + BAC - EV$ $= 2000 + 3000 - 1000$ $= 4000$	$EAC = BAC / CPI \text{ (CPI} = 1000 / 2000 \text{)}$ $= 3000 / .5$ $= 6000$

7. Which of the following group decision making techniques is used to reach a consensus on conflicting issue for the following situation?

Refer to the data below and answer the question.

Data given:

Total team size is 24 and includes project manager. When this group takes a decision, 11 out of 24 agree for the outcome.

- A. Unanimity
- B. Majority
- C. Dictatorship
- D. Plurality

Explanation:

Choice D is the correct answer.

Since 11 members agree, $11/24 = 45.83\%$.

So as per the definition, plurality is said when the largest block of a group decides outcome, even if a majority (that is more than 50%) is not achieved.

8. The initial investment on a pharmaceutical project is \$360000 with expected cash inflows of \$30000 per quarter, every quarter for the first three years and \$45000 per quarter from then on. What is the length of the time, an organization will take to recover the initial cost of the pharmaceutical project? The rate of return is 12%.
- A. 2
 - B. 3
 - C. 4
 - D. 5

Explanation:

Choice B is the correct answer.

The payback period is defined as the length of time period it takes performing organization to recover the initial investment costs of producing the product.

Total cash flow year 1, year 2 and year 3 = \$360000

The payback is reached in three years.

The fact that inflows are \$45000 per quarter starting in year 4 makes no difference because payback is already reached in three years.

9. The Project "I" has a NPV of \$175000 with an internal rate of return of 2.5% and an initial investment of \$115000.

Project J has a NPV of \$160000 with a BCR (Benefit to Cost Ratio) of 2.5:1.

Project K has a NPV of \$180000 which includes an opportunity cost of \$135000.

Based on these projects, which is the BEST one to select:

- A. Project I
- B. Project J
- C. Project K
- D. Some more information is needed to do project selection

Explanation:

Choice C is the correct answer.

The data have been given for the net present value (NPV) for each project. NPV means the total value that particular project is worth to organization. You need to select the project with the highest NPV.

10. A project manager has hired the services of a quality control specialist to determine the random variation in diameter of bearings being supplied by a vendor.

The quality control specialist evaluates the process at the vendor's work place and determines that the standard deviation due to the production process is 0.3 cm, and the measurement process is 0.05 cm. The specialist assumes that there are no other significant sources of variations.

What should be the standard deviation reported as?

- A. 0.350 cm
- B. 0.123 cm
- C. 0.063 cm
- D. 0.304 cm

Explanation:

Choice D is the correct answer.

The resultant standard deviation due to sources of variation is derived from variances due to each source.

The variances are additive. $2 \text{ Variance} = (\text{Sigma}) * (\text{Sigma}) = (\text{Standard Deviation})^2$

Variance due to production process = $0.3 * 0.3 = 0.09$

Variance due to measurement process = $0.05 * 0.05 = 0.0025$

Total variance = 0.0925

Therefore, Net standard variation = Square root of 0.0925 = 0.304 cm.

11. What would a CPI greater than 1 mean?

- A. Budget over run
- B. Schedule slippage
- C. Over performing on schedule
- D. Actual costs are less than earned value

Explanation:

Choice D is the correct answer. $\text{CPI} = \text{EV} / \text{AC}$ which if greater than 1 would imply that the actual costs are less than earned value. ($\text{AC} < \text{EV}$).

12. You accept the project costs to date and assume future work (ETC = Estimate to Complete) will be performed at the budgeted rate.

If BAC = 950, ETC = 400, PV = 475, AC = 325, EV = 425, and cumulative CPI = 1.31, what is the EAC?

- A. 740
- B. 950
- C. 600
- D. 850

Explanation:

Choice D is the correct answer. When you accept project performance to date and assume future ETC work will be performed at the budgeted rate, EAC is calculated as follows:

EAC = Actual costs incurred + Remaining budget as of today

Therefore, the EAC calculation for this question looks like this:

$$= (325) + (950 - 425)$$

13. Your customer has asked you for a forecast for the cost of the project completion, following project past performance data are available to you:

- 1) Total budget = \$70,000
- 2) CPI = 0.78
- 3) Actual expenditure till date = \$30,500

How much more money do you plan to spend on the project?

- A. \$70000
- B. \$89743.59
- C. \$59243.59
- D. \$107743.59

Explanation:

Choice C is the correct answer.

$$ETC = EAC - AC$$

Where

$$EAC = BAC / CPI \text{ (based on current rate)}$$

$$EAC = 70,000 / 0.78$$

= 89743.59. This how much total money you are going to spend for whole project...

$$\text{Now } ETC = EAC - AC$$

$$= 89743.59 - 30500$$

$$= 59243.59$$

14. A project manager has derived the "Activity Cost Estimates" for the project as \$ 250000. While preparing the risk response plans he realizes that he requires \$ 50000 as a contingency Reserves.

What is the cost baseline in this case?

- A. Same as cost budget
- B. 300000
- C. 200000
- D. The given information is inadequate

Explanation:

Choice B is the correct answer.

Cost baseline=Activity Cost Estimates + Contingency reserves

Cost Performance baseline = \$ 250000 + \$ 50000

= \$ 300000

15. If the upper specification limit and the lower specification limit of acceptance are at +/- 3 sigma, respectively, of the process capability, what % of the process outputs is likely to be rejected?

- A. 0.0014
- B. 0.0027
- C. 0.0454
- D. 0.0227

Explanation:

Correct answer is option B.

For a normal distribution process 99.73% of output would fall within the +/-3 sigma limits.

Therefore, percentage of outputs that would be outside the limit is $(100-99.73) \% = 0.27\%$.

16. The sponsor has issued a project charter to a project manager.

A summary budget of \$50000 has been allocated to this project while initiating it. Being an expert in project management area, you have provided an activity cost estimate as: \$25000 - \$75000.

This is an example of?

- A. Final Estimate

- B. Middle Estimate
- C. Rough Order of Magnitude Estimate
- D. Budgetary Estimate

Explanation:

Choice C is the correct answer.

A ROM (Rough order of magnitude) estimate is given at the very beginning of any project and is defined as +/- 50% of the original estimated cost. As per data given in the example, +/- 50% of \$50000 will be \$25000 and \$75000 respectively.

17. Assume that a project has only two activities A and B.

A project manager has used PERT technique. Activity A has optimistic estimates of 10 weeks, the pessimistic estimates of 15 weeks and the most likely estimates of 12 weeks. Activity B has optimistic estimates of 9 weeks, pessimistic estimates of 13 weeks and the most likely estimates of 11 weeks. What is the standard deviation?

- A. 1.07
- B. 1.14
- C. 0.83
- D. 0.92

Explanation:

Choice A is the correct answer.

18. An externally imposed project completion time necessitates the project duration as 75 days. The critical path of the project is established as 75 days, with a standard deviation of 3 days.

What could be the maximum project float available?

- A. Six days
- B. Zero days
- C. Three days
- D. Cannot be established

Explanation:

Choice C is the correct answer.

Project float is the difference between externally imposed completion period and estimated project duration, which is equal to the length of the critical path. The difference is zero. Add to this the

standard deviation of 3 days (implying that the critical path could be +/- 3 days from the estimated one). The maximum float could be $0+3 = 3$ (three) days.

19. A new project was initiated and two stakeholders were involved initially. Gradually the number of stakeholders increases to 7.

How many channels the project manager is managing now and how many additional channels got added due to increase in number of stakeholders in the project?

- A. 14 channels, 21 channels
- B. 7 channels, 7 channels
- C. 21 channels, 20 channels
- D. 21 channels, 7 channels

Explanation:

Choice C is the correct answer.

The correct answer is 21. The number of channels = $n(n-1)/2$ where n is the number of stakeholders. Since there are 7 stakeholders now, the number of channels is $7 \times 6 / 2 = 21$

To calculate how many additional channels got added, calculate no. of channels existed initially = $2(2-1)/2 = 1$.

New channels added due to increase in stakeholders size = $21 - 1 = 20$.

20. While analyzing already identified risks, you try to apply analytical techniques.

You analyze that the event Probability Profit of \$ 5 billion 70 percent and Loss of \$ 3 billion 30 percent.

What would be the expected monetary value of the project?

- A. \$ 4.5 billion Profit
- B. \$ 2.6 billion Profit
- C. \$ 4.4 billion
- D. Information not sufficient

Explanation:

Choice B is the correct answer.

Event Probability Calculation:

Expected value Profit of \$ 5 billion 70 percent = $P \times I = 0.7 \times 5 = \$ 3.5$ billion

Loss of \$ 3 billion 30 percent = $P \times I = 0.3 \times 3 = \$ 0.9$ billion

Expected monetary value = \$ 3.5 billion - \$ 0.9 billion = \$ 2.6 billion profit.

21. You have recently joined as a project manager in an aircraft manufacturing company. You have been given following data about sigma values for project quality management.

1 Sigma = 68.25%

2 Sigma = 95.46%

3 Sigma = 99.73%

6 Sigma = 99.999%

As per organization policy, you apply Six Sigma, however your test results show that out of 1 million tests, on an average, only four planes fell out of the sky. But still it is not a good number even to advertise to your customers. Since you are working on a product that where human lives are concerned.

What is the best course of action you decide further?

- A. Continue using Six Sigma as that is the best accuracy you can achieve.
- B. Lead the aircraft testing towards zero defect goal as six sigma in this case may not always accurate as human life is involved.
- C. Lead the aircraft testing towards three sigma goal
- D. Discuss with experienced quality manager further as you are not able to decide on your own

Explanation:

Choice B is the correct answer.

If you're conducting quality tests for aircraft reliability, Six Sigma results could mean that out of 1 million tests, on an average, only four planes fell out of the sky.

22. An Instrument measuring thickness of a screw of 18 mm diameter gives the following readings 19.01 mm, 19.00015 mm, 19.0002 mm and 18.900 mm. Assuming the instrument is capable of measuring up to the third place decimal, which of the following statement is correct

- A. The instrument is precise but not accurate
- B. The instrument is not precise but accurate
- C. The instrument is neither precise nor accurate
- D. The instrument is both precise and accurate

Explanation:

Choice A is the correct answer.

The instrument is not accurate but precise. Precision means the values of repeated measurements are clustered and have little scatter. As the name suggest, accuracy means how close the measured values are to the actual value. The measurements though precise are not accurate.

23. Which of the following could be an appropriate WBS code for a work package at the fourth level in a WBS where the top level code is 2.0

- A. 4.1
- B. 2.1.1.2
- C. 2.1.4
- D. 2.4

Explanation:

Choice B is the correct answer.

The answer is 2.1.1.2 since the question talks about the fourth level in the WBS, we need to have four levels (any number) separated by separator (dot).

24. Basis the data given. PV = 47000, AC = 43000, EV = 48000, EAC = 50000, and BAC = 52500. Calculate VAC of the project?

- A. 7000
- B. 2000
- C. 2500
- D. 3000

Explanation:

Choice C is the correct answer.

VAC is calculated this way: $VAC = BAC - EAC$.

Therefore, $52500 - 50000 = 2500$.

25. Your selection committee is debating between two projects. Project A has a payback period of 21 months. Project B has a cost of \$175,000, with expected cash inflows of \$75,000 the first year and \$25,000 per quarter after that. Which project should you recommend?

- A. Either Project A or Project B, because the payback periods are equal
- B. Project A, because Project B's payback period is 24 months
- C. Project A, because Project B's payback period is 26 months
- D. Project A, because Project B's payback period is 20 months

Explanation:

Choice B is the correct answer.

Project B has a payback period of 24 months; \$75,000 is received in the first 12 months, with another \$100,000 coming in over each of the next 4 quarters, or 1 year.

26. You have 9 key stakeholders to communicate with on your project. Which of the following is true?
- A. There are total 45 channels of communication, and is considered when using the communications requirements analysis tool and technique
 - B. There are total 45 channels of communication, and is considered when using the communications technology tool and technique
 - C. There are 36 channels of communication, and is considered when using the communications technology tool and technique
 - D. There are 36 channels of communication, and is considered when using the communications requirements analysis tool and technique

Explanation:

Choice A is the correct answer.

There are $[n*(n-1)]/2 = 10*9/2 = 45$ channels of communication, which is considered when using the communications requirements analysis tool and technique.

27. You are the project manager, considering recommending a project that costs \$520,000; expected inflows are \$40,000 per quarter for the first two years and then \$50,000 per quarter thereafter. What is the payback period?
- A. 40 months
 - B. 38 months
 - C. 36 months
 - D. 41 months

Explanation:

Choice C is the correct answer.

Year 1 and 2 inflows are each \$160,000 for a total of \$320,000. Year 3 inflows \$50,000 per quarter so total of \$200,000. Add third year the total of \$520,000 is reached in three years which is equal to 36 months.

28. You have received the following estimates for a complex activity that is critical to the success of your project. The most likely estimate is \$40, the optimistic estimate is \$36, and the pessimistic estimate is \$54. What is the expected activity cost of this activity (rounded figure)?
- A. \$ 45
 - B. \$ 39

- C. \$ 42
- D. \$ 41

Explanation

Choice C is the correct answer.

The expected activity cost is calculated this way: $(4 * \text{most likely} + \text{optimistic} + \text{pessimistic}) / 6$.
Therefore, the answer is $(4 * 40 + 36 + 54) / 6 = 42$.

29. A project is contracted as a cost-plus-fixed-fee (CPFF) contract, with a fee of 10% of estimated cost. The estimated cost is \$ 50,000. If the project comes in at \$75,000 with no change in project scope, what would be the total cost of contract?
- A. \$ 78,000
 - B. \$ 80,000
 - C. \$ 50,000
 - D. \$ 55,000

Explanation:

Choice B is the correct answer.

Target fee = $(10/100) * 50,000 = \$5,000$. Total price = cost + fee = $\$75,000 + \$5,000 = \$80,000$

30. A project is contracted as a cost-plus-incentive fee (CPIF) contract type. The project is negotiated such that if the final costs are less than expected cost, the sharing formula of cost savings is 75:25. The targeted cost is \$100,000 with an 8% incentive fee. If the project comes in at \$80,000, find the cost of total contract.
- A. \$108,000
 - B. \$ 80,000
 - C. \$ 93,000
 - D. \$ 88,000

Explanation:

Choice C is the correct answer.

Profit = Target Cost – Actual Cost = $100K - 80K = 20K$. Incentive fee is based on the budget costs = 8% of 100K = 8K. 25% of Profit = $(25/100) * 20K = 5K$
Total Cost = $80K + 8K + 5K = 93K$

31. The project manager accept project costs to date and assume future cost variances to be atypical. What is Estimate at completion (EAC) if
BAC=\$84500, ETC=\$30000, PV=\$42500, AC=\$40000, EV=\$45000 and cumulative CPI=1.35.

- A. \$82,500
- B. \$76,000
- C. \$79,500
- D. \$88,000

Explanation:

Choice c is the correct answer.

Use the formula $EAC = AC + BAC - EV$ in case of atypical variances

$$EAC = AC + BAC - EV$$

$$= 40000 + 84500 - 45000$$

$$= \$79,500.$$

32. If the benefit cost ratio of project A is 2.75, the benefit cost ratio of Project B is 1.5 and the benefit cost ratio of Project C is 1.95 which project would you select.

- A. Project A
- B. Project B
- C. Project C
- D. Information is not sufficient for project selection

Explanation

Choice A is the correct answer.

Project A – as the BCR of Project A is higher

33. A company records total sales of \$600,000 and carries a cash balance of \$7,000. Inventory turnover is \$150,000, and accounts receivable is \$40,000. What is the company's net working capital if accounts payable is \$52,000?

- A. \$120,000
- B. \$140,000
- C. \$145,000
- D. \$200,000

Explanation:

Choice C is the correct answer.

Net Working Capital = Current Assets – Current Liabilities

$$\text{Net Working Capital} = (\text{Cash} + \text{Inventory} + \text{A/R}) - (\text{A/P})$$

$$\Rightarrow \$7,000 + \$150,000 + \$40,000 - \$52,000 = \$145,000$$

34. If the expected value of an activity is 110 and the standard deviation is 12, which of the following is true?
- A. There is approximately a 99 percent chance of completing this activity in 90 to 124 days.
 - B. There is approximately a 68 percent chance of completing this activity in 98 to 122 days.
 - C. There is approximately a 95 percent chance of completing this activity in 86 to 122 days.
 - D. There is approximately a 75 percent chance of completing this activity in 90 to 124 days.

Explanation:

Choice B is the correct answer.

A 68 percent probability is calculated using plus or minus one standard deviation, a 95 percent probability uses plus or minus two standard deviations, and a 99 percent probability uses plus or minus three standard deviations.

35. You have two possible projects to manage, but you can only choose one. Project EMERALD is worth \$27,000, while Project PEARL is worth \$29,000. Management elects to choose Project PEARL. Which one of the following is the opportunity cost of this choice?

- A. \$27,000
- B. \$29,000
- C. \$23,000
- D. \$56,000

Explanation:

Choice A is the correct answer.

The opportunity cost is the amount of the project that was not chosen.

36. You are a project Manager in a software firm. There was a temporary need for a P3 server with a certain specification for your project. Basis the information given below, what will be your decision

- Rent - \$ 1250 Installation cost + \$250 rent per day
 - Buy - \$ 1250 installation cost + \$ 3500 onetime cost + \$75 per day Maintenance cost
- A. Buy if required for less than 20 days
 - B. Rent if required for more than 25 days
 - C. Buy if required for more than 20 days
 - D. Rent if required for more than 20 days

Explanation:

The answer is option C, Buy if required for more than 20 days

Let the Server is required for x days

Rent: Total=1250 + 250x

Buy: Total=1250 +3500 + 75x

Equate both to find the value of x

$$1250+250x= 1250 + 3500 + 75x$$

$$175x= 3500$$

$$x=20$$

For first 20 days it will cost equally for both buy and rent.

37. Three Team leaders of development submitted the individual cost estimates for their work packages as \$ 20,000, \$ 50,000 and \$ 35,000. The testing team submitted their estimate as \$ 30,000. You as the project manager had estimated the contingency reserve of \$15,000. The management reserve allocated to your project was \$ 10,000. Calculate the cost baseline of the project.

- A. \$ 135,000
- B. \$ 150,000
- C. \$ 160,000
- D. \$ 120,000

Explanation:

The answer is B

Cost Baseline = Project cost + Contingency Reserve

Project cost = Sum of control accounts

= Control Account1 (Development team) + Control Account2 (Testing Team)

= \$ 105,000 + \$ 30,000 = \$ 135,000

Cost Baseline = \$135,000 + \$15,000 =\$150,000

38. Scott was appointed as a project manager for a very critical project. He knows that variances that have occurred on the project to date are not expected to continue. Basis the data below calculate the ETC if
BAC=\$42,500, PV=\$40,000, AC=\$25,000 and Cumulative EV=\$32,500.

- A. \$ 10,000
- B. \$ 20,000
- C. \$ 7692.30
- D. \$ 7592.30

Explanation:

Choice A is the correct answer.

Use the formula of atypical variances

ETC=BAC- cum EV

= \$ 42500-\$32500

= \$10,000

39. If you know the expected value is 600 and the standard deviation is 14, you can say with approximately a 95 percent confidence rating which of the following?

- A. The activity will take from 586 to 614 days.
- B. The activity will take from 558 to 636 days.
- C. The activity will take from 568 to 606 days.
- D. The activity will take from 572 to 628 days.

Explanation:

Choice D is the correct answer.

There is a 95 percent probability that the work will finish within plus or minus two standard deviations. The expected value is 600, and 2 times standard deviation is 28 (i.e. 14×2), so the activity will take from 572 to 628 days.

40. From the data given below, which project risk should be addressed first?

Given that Low=\$10k, Medium=\$20K, High=\$35K

Project	Impact	Probability	Cost
A	Low	10%	\$20K
B	Medium	25%	\$28K
C	Low	30%	\$27K
D	High	20%	\$25K

- A. Project A
- B. Project B
- C. Project C
- D. Project D

Explanation

Choice B is the correct answer.

Project B – Applying the formula, the expected monetary value = cost +Probability*Impact
This value is highest for Project B

$$\text{Value} = \$28 + (20 \times 25\%) = \$33\text{K}$$

41. Basis data given below :

Project A is expected to make \$ 80,000 in two years.

Project B is expected to make \$100,000 in three years.

Project C is expected to make \$120,000 in five years.

If the cost of capital is 11 percent, which project should you choose?

- A. Project A
- B. Project B
- C. Project C
- D. Insufficient data to conclude anything

Explanation:

Choice B is the correct answer.

Using the formula to calculate present Value PV, calculate each project's worth:

$$\text{PV} = \text{FV} / (1 + \text{rate}/100)^{\text{year}}$$

The PV of Project A = \$ 64,930

The PV of Project B = \$ 73,120

The PV of Project C = \$ 71,214

Project B is the project that will return the higher investment to the company and should be chosen over Project A or C.

42. Consider the following data:

Activity	Duration (Days)	Predecessor Activity
A	8	-
B	6	A
C	2	B
D	10	B
E	9	C
F	6	B

What's the Critical Path duration?

- A. 30 Days
- B. 24 Days
- C. 25 Days
- D. 26 Days

Explanation:

Choice C is the correct answer.

Activity-D is on Critical path. If it is delayed by even a single day, it would delay the project by same duration

43. You are a project manager for a software development project. The final schedule was created and approved by all key stakeholders. Halfway through the execution phase, you have discovered that an activity on Critical path has been delayed by 2 days. What's the project float at this time?
- A. There is no project float.
 - B. It can't be determined from the given data.
 - C. There is a negative float of 2 days.
 - D. You have to apply crashing or fast tracking

Explanation:

Choice C is the correct answer.

Project float compares the project schedule against an externally imposed due date. The scenario explained would mean a negative project float of 2 days.

44. Which of the following is considered the most critical EVM metric?

- A. CV
- B. CPI
- C. SPI
- D. SP

Explanation:

Choice B is the correct answer.

CPI is considered the most critical EVM metric. It measures the cost efficiency of the project work completed at the measuring date.

45. Consider the following data:

Activity	Duration (Days)	Predecessor Activity
A	4	-
B	6	A
C	6	B
D	8	A
E	2	D
F	2	E
G	4	E
H	2	C

What's the Critical Path duration?

- A. 16 Days
 - B. 18 Days
 - C. 36 Days
 - D. 20 Days
46. As per the data in previous question, what's the float on activity C
- A. 2
 - B. 1
 - C. 0
 - D. -2
47. As per the data in question 45, what's the float on activity F
- A. 2
 - B. 1
 - C. 0
 - D. -2
48. Considering the schedule network diagram as per question 45, if management asks you to complete the project within 16 days, what's the total float for the project?
- A. 2
 - B. 1
 - C. 0
 - D. -2
49. What is the present value of \$100,000 to be received two years from now with an interest rate of 8%?
- A. 30864\$
 - B. 60096\$
 - C. 10000\$
 - D. 85734\$
50. After 25% of the scope is complete, you have found that 50,000\$ have already been spent from the total budget. If BAC is 200,000\$, then what's the CPI at this stage?
- A. 0.5
 - B. 0.75
 - C. 1.0
 - D. 1.25

Answers to questions 45 to 50

45. The correct answer is B.
There are two critical paths, A-B-C-H-End & A-D-E-G-End. Both have a duration of 18 Days
46. The correct answer is C.
Activity C is on Critical path. Float is zero
47. The correct answer is A.

It is evident from the diagram, you may also use the forward & backward pass method to find this out

48. The correct answer is D.

The duration of the project is 18 days at present. Asking to complete the project 2 days sooner, would mean a float of -2 days.

49. The correct answer is D.

$$NPV = FV / [1+r]^n = 100,000 / (1+0.08)^2 = 85,734$$

50. The correct answer is C.

If BAC is 200,000\$, then after 25% of the scope is complete, $EV = 25\% * 200,000\$ = 50,000\$$. As per the question, AC is also equal to 50,000\$. Hence $CPI = EV/AC = 1.0$

Detailed explanation of these questions and similar exclusive questions from other areas PMP are a part of MindCypress Virtual Instructor Led Training, Online Self Study and Simulator.

